

Mathematical modelling and application of frog choruses as an autonomous distributed communication system

Ikkyu Aihara, Daichi Kominami, Yasuharu Hirano and Masayuki Murata

Article citation details

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Review timeline

Original submission:	9 July 2018
1st revised submission:	16 October 2018
2nd revised submission:	25 November 2018
Final acceptance:	26 November 2018

Note: Reports are unedited and appear as submitted by the referee. The review history appears in chronological order.

Note: This manuscript was transferred from another Royal Society journal without peer review.

Review History

RSOS-181117.R0 (Original submission)

Review form: Reviewer 1 (G Ermentrout)

Is the manuscript scientifically sound in its present form?

Yes

Are the interpretations and conclusions justified by the results?

Yes

Is the language acceptable?

Yes

Is it clear how to access all supporting data?

Yes

Do you have any ethical concerns with this paper?

No

Have you any concerns about statistical analyses in this paper?

No

Recommendation?

Accept with minor revision (please list in comments)

Comments to the Author(s)

Overall this paper is most interesting when considering the frogs and less so than the wireless sensors. For the latter, I really think they need to compare it to some of the other methods. I have a number of questions about the numerical methods, but those are largely minor issues. There are some other points that need clarification and the paper needs to be edited by a native speaker.

I think you might also look at the papers by Belykh, Belykh and Hasler where synchrony was analyzed as the topology changed.

minor points (some less minor than others)

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page 7 In equation 2,3 you mean each time the phase hits $0 \bmod 2\pi$ the energy is decremented by 1 and the fatigue incremented by 1, correct? This should be stated explicitly since $\delta(x)$ means transversal crossing at $x=0$ and this is not the case here

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page 11 It seems like you synchronously evaluate the prob of switching every 3.5 seconds. This is a correct numerical implementation. I would think it should be more like a voltage gated ion channel where the rate of transition depends on the fatigue etc. this seems like a strange update rule. The right way would be to take the probability and divide by 3.5 to get a rate R and then if h is the time step for the ode model, at each such time step, choose a uniform random number, x , between 0 and 1. If $x < hR$ then make the transition. This is the proper approximation if you are assuming a cts markov process.

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So the θ_m inside Gamma makes no difference and this is the Winfree pulse coupled model. It would seem that this model would also be better for the frogs since they don't interact continuously but rather only when the other is calling. I understand that the cts interaction you guys show can come from weak coupling, but it would be better to be consistent in your modeling

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Instead of requiring the reader to figure out that $\sqrt{2}$ means talk to the 6 neighbors, just say it is a Moore neighborhood or say the 6 immediate neighbors

Eq15 is discontinuous at 0 and 2π . It should be continuous there like the sine model

Review form: Reviewer 2

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Yes

Is the language acceptable?

Yes

Is it clear how to access all supporting data?

Yes

Do you have any ethical concerns with this paper?

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Have you any concerns about statistical analyses in this paper?

No

Recommendation?

Major revision is needed (please make suggestions in comments)

Comments to the Author(s)

The authors investigated calling behavior of two or three frogs and found a group of frogs self-organizes their calling behavior in two different time scale; in short time scale, individual frog calls alternatively from the other frogs; in long time scale, a group of frogs switches between alternative calling state and silent state. The authors modeled these two behaviors using a coupled-oscillator system additionally introduced stochastic state switching mechanism. Last, the mechanism of this behavior was applied to a wireless sensor network problem.

Their findings of the two hierarchical behavior of the frog is interesting, and the idea of the application is fine. It is, however, the paper leaves room for improvement as listed below.

Main points

1. It is fine they succeeded in modeling of the frog behavior and found the silent state is useful for energy saving. So, how the efficiency depends on the parameters, especially on the coupling strength (K), the detection sensitivity of calling neighbors (H)? This kind of analysis would give a benefit in design of wireless sensor networks. The variable, energy E is not so important in the dynamics, if the author do not analyze the efficiency.
2. The proposed model is for just two or three frogs, while the number of units in the wireless sensor network is more than those up to 100. Why they did not construct a model of nearest (or second nearest) neighbor coupling system for the frog system (They did it in their previous work

ref[23]).

3. Most part of the basic idea has been already proposed in the reference [33] written by one of the authors of this submitted paper. This work should be introduced in the introduction part and the authors should present what is the main novel point of this paper differently from the previous one.

Minor points

1. The names of axes $G1(T_n)$ and $G2(T_n)$ in Fig. 3 are inconsistent with the explanation of Eq. (7,8) and Eq. (10); They are exchanged.
2. In Fig. 4B, last panel, it is difficult to find the feature of the tri-phase. How about the different combination of the data are colored different? The same problem is found in Fig. 7C, in which the partial magnified graph will be helpful for readers.

Decision letter (RSOS-181117.R0)

24-Sep-2018

Dear Dr Aihara,

The editors assigned to your paper ("Mathematical Modeling and Application of Frog Choruses as an Autonomous Distributed Communication System") have now received comments from reviewers. We would like you to revise your paper in accordance with the referee and Associate Editor suggestions which can be found below (not including confidential reports to the Editor). Please note this decision does not guarantee eventual acceptance.

Please submit a copy of your revised paper before 17-Oct-2018. Please note that the revision deadline will expire at 00.00am on this date. If we do not hear from you within this time then it will be assumed that the paper has been withdrawn. In exceptional circumstances, extensions may be possible if agreed with the Editorial Office in advance. We do not allow multiple rounds of revision so we urge you to make every effort to fully address all of the comments at this stage. If deemed necessary by the Editors, your manuscript will be sent back to one or more of the original reviewers for assessment. If the original reviewers are not available, we may invite new reviewers.

To revise your manuscript, log into <http://mc.manuscriptcentral.com/rsos> and enter your Author Centre, where you will find your manuscript title listed under "Manuscripts with Decisions." Under "Actions," click on "Create a Revision." Your manuscript number has been appended to denote a revision. Revise your manuscript and upload a new version through your Author Centre.

When submitting your revised manuscript, you must respond to the comments made by the referees and upload a file "Response to Referees" in "Section 6 - File Upload". Please use this to document how you have responded to the comments, and the adjustments you have made. In order to expedite the processing of the revised manuscript, please be as specific as possible in your response.

In addition to addressing all of the reviewers' and editor's comments please also ensure that your revised manuscript contains the following sections as appropriate before the reference list:

- Ethics statement (if applicable)

If your study uses humans or animals please include details of the ethical approval received, including the name of the committee that granted approval. For human studies please also detail whether informed consent was obtained. For field studies on animals please include details of all permissions, licences and/or approvals granted to carry out the fieldwork.

- Data accessibility

It is a condition of publication that all supporting data are made available either as supplementary information or preferably in a suitable permanent repository. The data accessibility section should state where the article's supporting data can be accessed. This section should also include details, where possible of where to access other relevant research materials such as statistical tools, protocols, software etc can be accessed. If the data have been deposited in an external repository this section should list the database, accession number and link to the DOI for all data from the article that have been made publicly available. Data sets that have been deposited in an external repository and have a DOI should also be appropriately cited in the manuscript and included in the reference list.

If you wish to submit your supporting data or code to Dryad (<http://datadryad.org/>), or modify your current submission to dryad, please use the following link:

<http://datadryad.org/submit?journalID=RSOS&manu=RSOS-181117>

- Competing interests

Please declare any financial or non-financial competing interests, or state that you have no competing interests.

- Authors' contributions

All submissions, other than those with a single author, must include an Authors' Contributions section which individually lists the specific contribution of each author. The list of Authors should meet all of the following criteria; 1) substantial contributions to conception and design, or acquisition of data, or analysis and interpretation of data; 2) drafting the article or revising it critically for important intellectual content; and 3) final approval of the version to be published.

All contributors who do not meet all of these criteria should be included in the acknowledgements.

We suggest the following format:

AB carried out the molecular lab work, participated in data analysis, carried out sequence alignments, participated in the design of the study and drafted the manuscript; CD carried out the statistical analyses; EF collected field data; GH conceived of the study, designed the study, coordinated the study and helped draft the manuscript. All authors gave final approval for publication.

- Acknowledgements

Please acknowledge anyone who contributed to the study but did not meet the authorship criteria.

- Funding statement

Please list the source of funding for each author.

Please note that Royal Society Open Science charge article processing charges for all new submissions that are accepted for publication. Charges will also apply to papers transferred to Royal Society Open Science from other Royal Society Publishing journals, as well as papers

submitted as part of our collaboration with the Royal Society of Chemistry (<http://rsos.royalsocietypublishing.org/chemistry>). If your manuscript is newly submitted and subsequently accepted for publication, you will be asked to pay the article processing charge, unless you request a waiver and this is approved by Royal Society Publishing. You can find out more about the charges at <http://rsos.royalsocietypublishing.org/page/charges>. Should you have any queries, please contact openscience@royalsociety.org.

Once again, thank you for submitting your manuscript to Royal Society Open Science and I look forward to receiving your revision. If you have any questions at all, please do not hesitate to get in touch.

Kind regards,
Royal Society Open Science Editorial Office
Royal Society Open Science
openscience@royalsociety.org

on behalf of Prof. Kevin Padian (Subject Editor)
openscience@royalsociety.org

Subject Editor's comments:

Thank you for your very interesting submission. Please be sure to address all the concerns of reviewers when you revise. Overall the English is very good but it would not hurt to have it looked over by a native speaker of English (a very difficult language!).

Associate Editor's comments:

Please fully respond to the concerns of the reviewers: if you do not do so to the satisfaction of the Editors and reviewers, we will be unable to accept further revisions.

Comments to Author:

Reviewers' Comments to Author:

Reviewer: 1

Comments to the Author(s)

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2. In Fig. 4B, last panel, it is difficult to find the feature of the tri-phase. How about the different combination of the data are colored different? The same problem is found in Fig. 7C, in which the partial magnified graph will be helpful for readers.

Author's Response to Decision Letter for (RSOS-181117.R0)

See Appendix A.

RSOS-181117.R1 (Revision)

Review form: Reviewer 1 (G Ermentrout)

Is the manuscript scientifically sound in its present form?

Yes

Are the interpretations and conclusions justified by the results?

Yes

Is the language acceptable?

Yes

Is it clear how to access all supporting data?

Yes

Do you have any ethical concerns with this paper?

No

Have you any concerns about statistical analyses in this paper?

No

Recommendation?

Accept as is

Comments to the Author(s)

Why is it called 1:2 synchronization? It remains 1:1 with each oscillator firing once per cycle. Also on page 4

If there are three frogs and two are synchronized, isn't there just 1 frog remaining? Instead of "rest pair", perhaps you mean remaining frog?

Review form: Reviewer 2

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Yes

Is it clear how to access all supporting data?

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Do you have any ethical concerns with this paper?

No

Have you any concerns about statistical analyses in this paper?

Yes

Recommendation?

Accept with minor revision (please list in comments)

Comments to the Author(s)

Most of my concerns have been addressed except for the followin minor points:

(Answer to my major comment 1)

The authors added a new measure t_{pow} to capture dependence of system behavior on parameters α , K_{nm} , and p_{high} . For K_{nm} , they examine on the domain $K_{\text{nm}} \geq 0.5$. How about $K_{\text{nm}} < 0.5$?

Relating to this, in page 10, line 258, they describe the parameter setting as $K_{\text{nm}} = 0.20$. In contrast they stated, in their rebuttal document, that "performed the simulation with the correct value (i.e., $K_{\text{nm}} = 1.0$). Please correct the conflict.

(Answer to my minor comment 2)

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(Page 7, line 184)

The authors termed the function as "Dirac delta function". It is, however, this is not accurate expression because $\delta = \infty$ when $\theta = 0$ in Dirac delta function. On the other hand their definition is $\delta = 1$ when $\theta = 0$, $\delta = 0$ otherwise.

Decision letter (RSOS-181117.R1)

13-Nov-2018

Dear Dr Aihara:

On behalf of the Editors, I am pleased to inform you that your Manuscript RSOS-181117.R1 entitled "Mathematical Modeling and Application of Frog Choruses as an Autonomous Distributed Communication System" has been accepted for publication in Royal Society Open Science subject to minor revision in accordance with the referee suggestions. Please find the referees' comments at the end of this email.

The reviewers and Subject Editor have recommended publication, but also suggest some minor revisions to your manuscript. Therefore, I invite you to respond to the comments and revise your manuscript.

- Ethics statement

If your study uses humans or animals please include details of the ethical approval received, including the name of the committee that granted approval. For human studies please also detail whether informed consent was obtained. For field studies on animals please include details of all permissions, licences and/or approvals granted to carry out the fieldwork.

- Data accessibility

It is a condition of publication that all supporting data are made available either as supplementary information or preferably in a suitable permanent repository. The data accessibility section should state where the article's supporting data can be accessed. This section should also include details, where possible of where to access other relevant research materials such as statistical tools, protocols, software etc can be accessed. If the data has been deposited in an external repository this section should list the database, accession number and link to the DOI for all data from the article that has been made publicly available. Data sets that have been deposited in an external repository and have a DOI should also be appropriately cited in the manuscript and included in the reference list.

If you wish to submit your supporting data or code to Dryad (<http://datadryad.org/>), or modify your current submission to dryad, please use the following link:
<http://datadryad.org/submit?journalID=RSOS&manu=RSOS-181117.R1>

- Competing interests

Please declare any financial or non-financial competing interests, or state that you have no competing interests.

- Authors' contributions

All submissions, other than those with a single author, must include an Authors' Contributions section which individually lists the specific contribution of each author. The list of Authors should meet all of the following criteria; 1) substantial contributions to conception and design, or acquisition of data, or analysis and interpretation of data; 2) drafting the article or revising it critically for important intellectual content; and 3) final approval of the version to be published.

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- Acknowledgements

Please acknowledge anyone who contributed to the study but did not meet the authorship criteria.

- Funding statement

Please list the source of funding for each author.

Please note that we cannot publish your manuscript without these end statements included. We have included a screenshot example of the end statements for reference. If you feel that a given heading is not relevant to your paper, please nevertheless include the heading and explicitly state that it is not relevant to your work.

Because the schedule for publication is very tight, it is a condition of publication that you submit the revised version of your manuscript before 22-Nov-2018. Please note that the revision deadline will expire at 00.00am on this date. If you do not think you will be able to meet this date please let me know immediately.

To revise your manuscript, log into <https://mc.manuscriptcentral.com/rsos> and enter your Author Centre, where you will find your manuscript title listed under "Manuscripts with Decisions". Under "Actions," click on "Create a Revision." You will be unable to make your revisions on the originally submitted version of the manuscript. Instead, revise your manuscript and upload a new version through your Author Centre.

When submitting your revised manuscript, you will be able to respond to the comments made by the referees and upload a file "Response to Referees" in "Section 6 - File Upload". You can use this to document any changes you make to the original manuscript. In order to expedite the processing of the revised manuscript, please be as specific as possible in your response to the referees.

When uploading your revised files please make sure that you have:

- 1) A text file of the manuscript (tex, txt, rtf, docx or doc), references, tables (including captions) and figure captions. Do not upload a PDF as your "Main Document".
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- 3) Included a 100 word media summary of your paper when requested at submission. Please ensure you have entered correct contact details (email, institution and telephone) in your user account
- 4) Included the raw data to support the claims made in your paper. You can either include your data as electronic supplementary material or upload to a repository and include the relevant doi within your manuscript
- 5) All supplementary materials accompanying an accepted article will be treated as in their final form. Note that the Royal Society will neither edit nor typeset supplementary material and it will be hosted as provided. Please ensure that the supplementary material includes the paper details where possible (authors, article title, journal name).

Supplementary files will be published alongside the paper on the journal website and posted on

the online figshare repository (<https://figshare.com>). The heading and legend provided for each supplementary file during the submission process will be used to create the figshare page, so please ensure these are accurate and informative so that your files can be found in searches. Files on figshare will be made available approximately one week before the accompanying article so that the supplementary material can be attributed a unique DOI.

Please note that Royal Society Open Science charge article processing charges for all new submissions that are accepted for publication. Charges will also apply to papers transferred to Royal Society Open Science from other Royal Society Publishing journals, as well as papers submitted as part of our collaboration with the Royal Society of Chemistry (<http://rsos.royalsocietypublishing.org/chemistry>). If your manuscript is newly submitted and subsequently accepted for publication, you will be asked to pay the article processing charge, unless you request a waiver and this is approved by Royal Society Publishing. You can find out more about the charges at <http://rsos.royalsocietypublishing.org/page/charges>. Should you have any queries, please contact openscience@royalsociety.org.

Once again, thank you for submitting your manuscript to Royal Society Open Science and I look forward to receiving your revision. If you have any questions at all, please do not hesitate to get in touch.

Kind regards,
Royal Society Open Science Editorial Office
Royal Society Open Science
openscience@royalsociety.org

on behalf of Professor Kevin Padian (Subject Editor)
openscience@royalsociety.org

Associate Editor Comments to Author:

Thank you for submitting this revision. The referees have reviewed it and have a number of minor requests for you to address. We look forward to receiving your further revision.

Reviewer comments to Author:

Reviewer: 1

Comments to the Author(s)

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Author's Response to Decision Letter for (RSOS-181117.R1)

See Appendix B.

Decision letter (RSOS-181117.R2)

26-Nov-2018

Dear Dr Aihara,

I am pleased to inform you that your manuscript entitled "Mathematical Modeling and Application of Frog Choruses as an Autonomous Distributed Communication System" is now accepted for publication in Royal Society Open Science.

You can expect to receive a proof of your article in the near future. Please contact the editorial office (openscience_proofs@royalsociety.org and openscience@royalsociety.org) to let us know if you are likely to be away from e-mail contact. Due to rapid publication and an extremely tight schedule, if comments are not received, your paper may experience a delay in publication.

Please note that the email address of your co-author Yasuharu Hirano (y-hirano@ist.osaka-u.ac.jp) is not receiving emails from ScholarOne, please ensure that you supply us with an alternative email.

Royal Society Open Science operates under a continuous publication model (<http://bit.ly/cpFAQ>). Your article will be published straight into the next open issue and this will be the final version of the paper. As such, it can be cited immediately by other researchers. As the issue version of your paper will be the only version to be published I would advise you to check your proofs thoroughly as changes cannot be made once the paper is published.

On behalf of the Editors of Royal Society Open Science, we look forward to your continued contributions to the Journal.

Kind regards,
Andrew Dunn
Royal Society Open Science Editorial Office
Royal Society Open Science
openscience@royalsociety.org

on behalf of Prof Kevin Padian (Subject Editor)
openscience@royalsociety.org

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Dear Prof. Kevin Padian and Associate Editor,

We are pleased to send you the revised version of our manuscript. We have made most of the changes and corrections suggested by the reviewers. In addition, we have improved English of our manuscript by using a professional English editing service of a native speaker according to the comment of the first reviewer. A marked-up copy of the changes from the previous manuscript is also attached as a different file. We appreciate all the comments and suggestions as they have greatly improved the quality of our study.

As described in the responses to both reviewers, we found one mistake about a parameter value. Namely, in the previous manuscript, it was stated that the parameter K_{nm} is fixed at 1.0 for the simulation of a wireless sensor network. However, we carried out the simulation with a wrong value of $K_{nm} = 0.5$. We are sorry for this mistake. For this revision, we have performed the simulation with the correct value (i.e., $K_{nm} = 1.0$) and then have updated Figures 7, 8 and 9. Subsequently, the figures look slightly different from the previous ones while the results and conclusions hold.

We would like to thank again for the valuable comments, and wish that the revised manuscript is now suitable for the publication in *Royal Society Open Science*.

Sincerely yours,

Ikkyu Aihara (for all the authors)

2018/10/17

Reviewers' Comments to Author:

Reviewer: 1

Comments to the Author(s):

Overall this paper is most interesting when considering the frogs and less so than the wireless sensors. For the latter, I really think they need to compare it to some of the other methods. I have a number of questions about the numerical methods, but those are largely minor issues. There are some other points that need clarification and the paper needs to be edited by a native speaker.

I think you might also look at the papers by Belykh, Belykh and Hasler where synchrony was analyzed as the topology changed.

Acknowledgement:

Thank you very much for your valuable comments. We updated the manuscript according to your comments as shown below. We also appreciate your suggestion on the check by a native speaker. We used a professional English editing service by a native speaker and modified English of our manuscript.

In addition to the following modifications, we found one mistake about a parameter value in the simulation of a wireless sensor network. Namely, in the previous manuscript, it was stated that the parameter K_{nm} is fixed at 1.0 for the simulation. However, we carried out the simulation with a wrong value of $K_{nm} = 0.5$. We are sorry for this mistake. For this revision, we have performed the simulation with the correct value (i.e., $K_{nm} = 1.0$) and then have updated Figures 7, 8 and 9 of our manuscript. Subsequently, the figures look slightly different from the previous ones while the results and conclusions hold.

Revised Point:

We appreciate your comment on the comparison of the proposed model to alternative methods of a wireless sensor network. The main features of the proposed model can be summarized as follows: (1) neighboring nodes can avoid packet collision over a short time scale by alternating the timing of data transmission, and (2) all the nodes collectively switch their states over a long time scale, establishing high network connectivity while reducing network power consumption. Although there are alternative methods on the avoidance of packet collision over a short time scale, we could not find the methods realizing the second feature over a long time scale unfortunately. Consequently, we consider that the comparison with alternative methods achieving both of the above features is now difficult. In the updated manuscript, we explained these discussions on the relevance of partially related methods over a short time scale as well as the novelty of the proposed model over a long time scale (Lines 480-486 in page 18 (Summary of the proposed model); Lines 487-491 in page 18 (The relevance with alternative methods over a short time scale); Lines 509-515 in page 19 (The novelty of the proposed model over a long time scale)).

We also thank you for the comment on the study by Belykh et al. We agree that their studies on the synchrony and network topology are related to our study. This is because the temporal structures emerging from anti-phase synchronization of neighboring oscillators are also likely to be

affected by the change of network topology. We explain this point in the section of Conclusions and Discussions with citing the studies of Belykh et al. (Lines 501-508 in page 19).

Minor points (some less minor than others)

Comment 1:

page 2 flock -> flocking

Revised Point:

We modified the word "flock" in the introduction to "flocking" according to your comment (Line 30 in page 2). In addition, we found the same word in the abstract, so we also modified it (Line 13 in page 1).

Comment 2:

page 3 to -> over

Revised Point:

Thank you for the comment. In page 3 of the previous manuscript, however, there were multiple "to" and we could not figure out which one should be modified. So, we ask a native speaker of a professional English editing service to check our manuscript. However, the native speaker did not point out that any of the multiple "to" in the page 3 should be modified. Hence, we could not modify the word in this revision.

Comment 3:

page 7 In equation 2,3 you mean each time the phase hits $0 \bmod 2\pi$ the energy is decremented by 1 and the fatigue incremental by 1, correct? This should be stated explicitly since $\delta(x)$ means transversal crossing at $x=0$ and this is not the case here

Revised Point:

Thank you for the comment. Yes, our model assumes that E_n is decremented by 1 and T_n is incremented by 1 each time the phase hits 0. Because this calculation is not the usual case of the Dirac delta function as you point out, we add the above explanation to our manuscript (Lines 190-193 in page 7).

Comment 4:

page 10 In your earlier PRE paper you used $\sin(x) + c \sin(2x)$ in order to get clusters as well as the triphase state. Is there a reason you don't use that in this paper since that seems to be what the frogs do

Revised Point:

Thank you for the comment. In the previous manuscript, we assumed the interaction term $K_{nm} \sin(\theta_n - \theta_m)$ as a simple function that can explain tri-phase synchronization of three frogs (Aihara et al., *Mathematical Biosciences*, 2008). However, 1:2 anti-phase synchronization of three frogs (i.e., two pairs of three frogs call alternately while the rest pair overlap their calls) is also observed in the chorus of actual male Japanese tree frogs as you point out (see Figure 1D), and then the interaction term of $K_{nm} [\sin(\theta_n - \theta_m) - k \sin(2(\theta_n - \theta_m))]$ can reproduce both 1:2 anti-phase synchronization and tri-phase synchronization at the same parameter values (Aihara et al., *Phy Rev E*, 2011). Hence, we agree that the latter interaction term including the second-order component is better for the model of frog choruses.

For this revision, we have redone all the simulations by using the interaction term including the second-order component (i.e., $K_{nm} [\sin(\theta_n - \theta_m) - k \sin(2(\theta_n - \theta_m))]$) on the assumption of the pulse coupling that you point out in Comment 7 (see Equations (1) and (13)). Then, we have updated Figures 4 and 5. Consequently, the tri-phase synchronization and the 1:2 anti-phase synchronization occur at a different chorus in the same simulation (see the bottom panel of Figure 4B and also Figure S1 in Supplementary Information). We consider that these results reproduce the temporal patterns of frog choruses much more precisely. Thank you very much again for the valuable comment that has greatly improved the quality of our study.

Comment 5:

page 11 It seems like you synchronously evaluate the prob of switching every 3.5 seconds. This is a correct numerical implementation. I would think it should be more like a voltage gated ion channel where the rate of transition depends on the fatigue etc. this seems like a strange update rule. The right way would be to take the probability and divide by 3.5 to get a rate R and then if h is the time step for the ode model, at each such time step, choose a uniform random number, x , between 0 and 1. If $x < h R$ then make the transition. This is the proper approximation if you are assuming a cts markov process.

Revised Point:

Thank you for the comment. As you point out, we evaluate the probability of the transition every 3.5 seconds. In other words, we treat the transition as a discrete-time Markov process. The reason of this treatment is the discontinuity in the probability of the transition. As shown in Equations (7) and (8), the probability of the transition from the calling state to the silent state depends on the fatigue (T_n). Then, the time evolution of T_n during the calling state is described by the Dirac delta function (Equation (2)). Therefore, the fatigue T_n changes discretely in time, and then the probability of the transition also changes discretely in time.

To evaluate the probability of the transition (P) as a continuous-time Markov process, we recognize that the following steps are necessary: (1) we calculate the time differential of the probability of the transition (P'), and (2) we update the state according to the rate of the transition (i.e., $P' * h$) by using the time step h of the simulation. In this study, however, the probability of the transition is discontinuous in time as explained above, so that we cannot calculate the time

differential of the probability (i.e., P'). Subsequently, we treat the transition as a discrete-time Markov process which is an alternative approach of a continuous-time Markov process. We add these explanations to the manuscript (Lines 293-300 in pages 11 and 12).

Comment 6:

page 12

When the animal switches to the call state is the phase the same as it was when it left the call state in the previous bout? This seems unrealistic that it would recall the phase. Seems more likely it starts randomly or at 0

Revised Point:

Thanks for the comment. Actually, in the simulation of the previous manuscript, we randomized the phase from 0 to 2π at each transition from the silent state to the calling state. We are sorry that the manuscript lacked this information. We add the explanation about the randomization of the phase to the manuscript (Lines 305-306 in page 12).

Comment 7:

page 14

So the θ_m inside Gamma makes no difference and this is the Winfree pulse coupled model. It would seem that this model would also be better for the frogs since they don't interact continuously but rather only when the other is calling. I understand that the cts interaction you guys show can come from weak coupling, but it would be better to be consistent in your modeling

Revised Point:

We agree with your comment. As you point out, the calls of male Japanese tree frogs have a specific duration (Figure 1), indicating that they do not continuously interact. For this revision, we carried out the simulation by assuming the Dirac delta function also in the model of frog choruses (Equation (1)) and updated Figures 4 and 5. Consequently, we succeeded in reproducing the alternating chorus patterns of male Japanese tree frogs as in the case of the continuous interaction. Note that in the updated simulation we assume a slightly weaker interaction ($K_{nm} = 0.2$) than the previous case of the continuous interaction ($K_{nm} = 1.0$) so as to obtain the consistent result from the simulation.

Comment 8:

page 15

Instead of requiring the reader to figure out that $\sqrt{2}$ means talk to the 6 neighbors, just say it is a Moore neighborhood or say the 6 immediate neighbors

Revised Point:

We agree. In the updated manuscript, we first explain that we assume the situation in which each node interacts with a Moore neighborhood, and then provide an additional information explaining that such an interaction can be realized by a specific parameter value (Lines 377-379 in pages14 and 15).

Comment 9:

Eq15 is discontinuous at 0 and 2π . It should be cts there like the sine model

Revised Point:

Thank you for the comment. As you point out, we use a linear interaction term (Equation (14) that corresponds to Equation (15) in the previous manuscript) that is discontinuous at 0 and 2π . This is because it is required to avoid a packet collision (i.e., in-phase synchronization) as quickly as possible in a wireless sensor network. Although the sinusoidal interaction term can make the in-phase synchronization to be unstable, it takes much long time for the nodes to avoid the in-phase synchronization because of a very weak interaction near the in-phase synchronization (see Figure S2A in Supplementary Information). By contrast, the linear interaction term produces a strong interaction even at the in-phase synchronization: namely, the linear term takes a large positive value at 0 and takes a large negative value at 2π (see Figure S2B in Supplementary Information), which allows sensor nodes to avoid the in-phase synchronization very quickly. Therefore, we use the linear interaction term for the model of a wireless sensor network in this study although such a discontinuous function is not usually utilized for the model of a continuous-time system. We add this explanation to our manuscript (Lines 387-395 in page 15).

Reviewer: 2**Comments to the Author(s):**

The authors investigated calling behavior of two or three frogs and found a group of frogs self-organizes their calling behavior in two different time scale; in short time scale, individual frog calls alternatively from the other frogs; in long time scale, a group of frogs switches between alternative calling state and silent state. The authors modeled these two behaviors using a coupled-oscillator system additionally introduced stochastic state switching mechanism. Last, the mechanism of this behavior was applied to a wireless sensor network problem.

Their findings of the two hierarchical behavior of the frog is interesting, and the idea of the application is fine. It is, however, the paper leaves room for improvement as listed below.

Acknowledgement:

Thank you very much for your valuable comments. We updated the manuscript according to your comments as follows. The dynamical behaviors of various animals attract a great deal of attention from many researchers, and various experimental and theoretical studies were carried out. To our knowledge, however, the hierarchical feature of the animal behavior was not studied well especially from the viewpoint of mathematical modeling. We hope that this study will highlight the unique behavior of male frogs and also inspire further studies on bio-inspired technologies.

In addition to the following modifications, we found one mistake about a parameter value in the previous manuscript. Namely, in the simulation of a wireless sensor network, it was stated that the parameter K_{nm} is fixed at 1.0. However, we carried out the simulation with a wrong value of $K_{nm} = 0.5$. We are sorry for this mistake. For this revision, we have performed the simulation with the correct value (i.e., $K_{nm} = 1.0$) and then have updated Figures 7, 8 and 9 of our manuscript. Consequently, the figures look slightly different from the previous ones while the results and conclusions hold.

Main points**Comment 1:**

It is fine they succeeded in modeling of the frog behavior and found the silent state is useful for energy saving. So, how the efficiency depends on the parameters, especially on the coupling strength (K), the detection sensitivity of calling neighbors (H)? This kind of analysis would give a benefit in design of wireless sensor networks. The variable, energy E is not so important in the dynamics, if the author do not analyze the efficiency.

Revised Point:

Thank you very much for the valuable comment. We agree with you. To evaluate the efficacy of energy saving in a wireless sensor network, we need to examine how the dynamics of E_n is affected by the parameters of our model. Hence, we newly define the pace of power consumption (t_{pow}) and then examine how t_{pow} depends on the parameters of our model (Note that t_{pow} is defined as the time in which the remaining power of all the nodes becomes less than 90% of their

maximum power $E_{\max}$ (Lines 442-447 in page 17)). Consequently, we found that the parameter α in Equation (5) dominantly affects the efficacy of energy saving.

Figure 9B shows the relationship between α and t_{pow} . This figure demonstrates that it takes a longer time for sensor nodes to consume their energy if the parameter α takes a smaller value, indicating that we can control network lifetime by the parameter α . The origin of this result is that α represents the recovery rate of T_n (Equation (5)) and therefore determines the duration of the sleep state. We added these new analysis and result to our manuscript (Lines 442-451 in page 17).

In addition, we examine how t_{pow} depends on the parameters K and p_{high} according to your comment (Note that p_{high} is the parameter included in the function H (see Equation (12)) and represents the sensitivity of response to the signals from other nodes). The results of these analyses are shown in Figures S4 in Supplementary Information. In both figures, the range of the vertical axis is set at the same value with Figure 9B. It is demonstrated that these two parameters do not strongly affect t_{pow} compared to the parameter α . We guess that the reasons of these results are as follows: (1) the parameter K just slightly modulates an inter-signal interval of an active node and therefore does not have a strong effect on the duration of the active state, and (2) the parameter p_{high} mainly determines the occurrence of the collective transition but does not modulate the duration of each state.

Comment 2:

The proposed model is for just two or three frogs, while the number of units in the wireless sensor network is more than those up to 100. Why they did not construct a model of nearest (or second nearest) neighbor coupling system for the frog system (They did it in their previous work ref[23]).

Revised Point:

Thank you for the comment. In the section of frog choruses, this study aims to propose a mathematical model that can reproduce the features of empirical data over multiple time scales. To precisely examine the validity of the proposed model, we need larger datasets of empirical data. The reason of our restriction on the number of frogs (i.e., two frogs or three frogs) is the difference of the data size between indoor experiments and field observations. Namely, the data of our indoor experiments consists of the audio recordings of 16 hours (4 hours * 4 datasets; Aihara et al., *Phys Rev E*, 2011) while the data of our field observations consist of the video recordings of 1.25 hours (15 min * 5 datasets; Aihara et al., *Sci Rep*, 2014). Therefore, the datasets of field observations are currently much smaller than those of indoor experiments, and then we focus on the system of two frogs or three frogs in this manuscript so as to utilize the larger datasets of the indoor experiments. We add this explanation to our manuscript (Lines 466-472 in page18).

On the other hand, it remains as a future problem to extend the proposed model to a system of many frogs. Especially, in our previous study (Aihara et al., *Sci Rep*, 2014), we consider the interaction that decays depending on the distance among frogs; such a network topology should affect the temporal structures also in the proposed model. It is very interesting for us to examine the relationship between the network topology and the temporal chorus pattern on the basis of

such an extended model with many frogs. In regard to this point, it is also an important future problem to investigate the relationship between the network topology and the temporal structures in a wireless sensor network because we here consider specific network topology shown in Figure 7A (i.e., a two-dimensional square lattice in which each node interacts with a Moore neighborhood) although such a network topology should vary a lot in the practical usage of a wireless sensor network. We also add these discussions to our manuscript (Lines 473-479 in page 18; Lines 501-508 in page 19).

Comment 3:

Most part of the basic idea has been already proposed in the reference [33] written by one of the authors of this submitted paper. This work should be introduced in the introduction part and the authors should present what is the main novel point of this paper differently from the previous one.

Revised Point:

Thank you very much for the valuable comment. As you point out, Mutazono et al. carried out a seminal work on the application of frog choruses to a wireless sensor network (Mutazono et al., *Computer Communications*, 2012). We agree that their study should be cited in the introduction and then the novel point of our study should be explained.

As for the dynamics over a short time scale, Mutazono et al. also use a phase oscillator model so as to avoid a packet collision among sensor nodes. Namely, they propose a phase oscillator model with citing our theoretical study on frog choruses (Aihara et al., *Artificial Life and Robotics*, 2008), and then apply the model to the control of a wireless sensor network. Therefore, the study of Mutazono et al. is the first one that reports the application of the short-time-scale dynamics of frog choruses (i.e., anti-phase synchronization of two frogs) to the control of a wireless sensor network.

By contrast, we consider that this study proposes a novel approach on the dynamics over a long time scale. The point is that Mutazono et al. utilize *the satellite behavior* of male frogs to reduce network power consumption. The satellite behavior is the behavior that male frogs stay silent in the vicinity of a calling male for a tremendously long time in order to intercept a female that is attracted to the calling male (Gerhardt and Huber 2002; Wells 2007), which can be understood as *an inhibitory interaction* among male frogs that inhibits the calling behavior of neighboring males. In this study, we do not assume the satellite behavior of male frogs at all for the modeling of frog choruses. By contrast, we here focus on the collective and cyclic transition between the silent state and the calling state, and model the transition by using *an excitatory interaction* among male frogs that activate the calling behavior of neighboring males (Equation (12)). Because Mutazono et al. do not focus on the collective and cyclic transition and then do not reproduce it by their model, we consider that this study provides a novel approach for the dynamics over a long time scale.

We add these explanations to the introduction (Lines 72-85 in page 3).

Minor points

Comment 1:

The names of axes $G1(T_n)$ and $G2(T_n)$ in Fig. 3 are inconsistent with the explanation of Eq. (7,8) and Eq. (10); They are exchanged.

Revised Point:

Thank you very much for the comment. As you point out, $G1(T_n)$ and $G2(T_n)$ in Figure 3 of the previous manuscript should be $G2(T_n)$ and $G1(T_n)$, respectively. We are so sorry for this mistake. We modified the names of the axes according to your comment (see Figure 3 in the updated manuscript).

Comment 2:

In Fig. 4B, last panel, it is difficult to find the feature of the tri-phase. How about the different combination of the data are colored different? The same problem is found in Fig. 7C, in which the partial magnified graph will be helpful for readers.

Revised Point:

We agree with you. According to the comment, we update the bottom panel of Figure 4B by using color plots in which pink dots, blue dots and green dots represent θ_1 - θ_2 , θ_1 - θ_3 and θ_2 - θ_3 , respectively. Then, we also add the enlarged version of this figure as Supplementary Information so as to clearly show the details of the phase dynamics (see Figure S1 in Supplementary Information).

We also appreciate for your comment on Figure 7C. We show the enlarged version of Figure 7C in Supplementary Information (see Figure S3). We agree that this figure is helpful for readers. However, we could not find a specific temporal pattern in the phase dynamics. The reason of this complicated pattern would be the frustration emerging from anti-phase synchronization of neighboring oscillators. In our previous study (Aihara et al., *Sci. Rep.*, 2014), we showed that several synchronization states (i.e., two-cluster synchronization and wavy anti-phase synchronization with various wave numbers) are multi-stable even in a simple spatial structure of a ring when assuming oscillators that attempt to synchronize in anti-phase with neighbors. The point is that in this study we assume a more complicated spatial structure (i.e., a two-dimensional square lattice in which each oscillator interacts with a Moore neighborhood) than a ring. Therefore, a more complicated pattern is likely to emerge in this simulation. We consider that further analysis is required to reveal the details of the complicated pattern. We add this explanation to our manuscript (Lines 491-501 in pages 18 and 19).

Appendix B

Reviewer comments to Author

Review 1

Comment

Why is it called 1:2 synchronization? It remains 1:1 with each oscillator firing once per cycle.

Also on page 4

If there are three frogs and two are synchronized, isn't there just 1 frog remaining? Instead of "rest pair", perhaps you mean remaining frog?

Response:

Thank you very much for the comment. 1:2 anti-phase synchronization means that one frog synchronizes in anti-phase with the remaining two frogs that synchronize in in-phase (as you point out, this is the same with the state in which two frogs synchronize in in-phase, and they synchronize in anti-phase with the remaining frog). In the previous manuscript, we explained this situation based on the synchronized state of each pair, but we consider that the explanation was very confusing. Therefore, we modified the explanation of 1:2 anti-phase synchronization according to your comment ([Line 105 in Page 4](#)).

In addition, the phrase of 1:2 anti-phase synchronization seems to be also confusing for describing this state because each oscillator fires once per cycle as you point out. Therefore, we modified the phrase of 1:2 anti-phase synchronization to clustered anti-phase synchronization ([Line 106 in Page 4, Line 323 in Page 12, the caption of Figure 1, the caption of Figure 4, and the caption of Figure S1 in Supplementary Information](#)) so as to precisely describe the situation that one frog synchronizes in anti-phase with the remaining two frogs that synchronize in in-phase. Thank you very much again for your valuable comment.

Reviewer comments to Author

Review 2

General Comment

Most of my concerns have been addressed except for the followin minor points:

Acknowledgement:

Thank you very much for your valuable comments. We revised our manuscript as follows.

Comment 1

(Answer to my major comment 1)

The authors added a new measure t_{pow} to capture dependence of system behavior on parameters α , K_{nm} , and p_{high} . For K_{nm} , they examine on the domain $K_{nm} \geq 0.5$. How about $K_{nm} < 0.5$? Relating to this, in page 10, line 258, they describe the parameter setting as $K_{nm} = 0.20$. In contrast they stated, in their rebuttal document, that “performed the simulation with the correct value (i.e., $K_{nm} = 1.0$). Please correct the conflict.

Response:

Thank you for the comment. In this study, we use different types of interaction terms as well as their coupling strengths for the simulations of frog choruses and a wireless sensor network. For the simulation of frog choruses, we use a sinusoidal interaction term of Equation (13) with the coupling strength of $K_{nm} = 0.2$ ([Line 265 in Page 10 in the current manuscript](#)). In contrast, for the simulation of a wireless sensor network, we use a linear interaction term of Equation (14) with the coupling strength of $K_{nm} = 1.0$ ([Line 404 in Page 15](#)). In the rebuttal document of our previous submission, we stated our previous mistake about the simulation of [a wireless sensor network](#) as follows:

Rebuttal document of our previous submission

“ In addition to the following modifications, we found one mistake about a parameter value in the previous manuscript. Namely, in the simulation of [a wireless sensor network](#), it was stated that the parameter K_{nm} is fixed at 1.0. However, we carried out the simulation with a wrong value of $K_{nm} = 0.5$. We are sorry for this mistake. For this revision, we have performed the simulation with the correct value (i.e., $K_{nm} = 1.0$) and then have updated Figures 7, 8 and 9 of our manuscript. Consequently, the figures look slightly different from the previous ones while the results and conclusions hold.”

This statement explains that we used a correct value for the simulation of frog choruses (i.e., $K_{nm} = 0.2$; [Line 265 in Page 10](#)) but used a wrong value for the simulation of a wireless sensor network (i.e., we used $K_{nm} = 0.5$ although the correct value is $K_{nm} = 1.0$ as explained in [Line 404 in Page 15](#)). We are sorry for the confusing explanation about the parameter values. We are sure that all the simulations have been carried out with the correct values in the current manuscript.

On the other hand, we agree that the simulation in a wider parameter range is quite important. In this revision, we carried out the same simulation in a wider range of $0 \leq K_{nm} < 1.5$ that includes the range of $K_{nm} < 0.5$. According to the simulation, we updated Figure S4A in the Supplementary Information. Thank you very much again for the valuable comment.

Comment 2

(Answer to my minor comment 2)

They added an enlarged graph of the last panel of Fig.4B into Fig. S1. The both graphs are identical then readers need not duplication. My comment in the previous report meant that it will be helpful if a part of the graph, for example, that of time= 8650-8800, is magnified to be able to capture the process toward phase locking.

Response:

Thank you very much for the comment. We are sorry that we misunderstood your comment in our rebuttal document of the previous submission. We agree that the graph enlarged in the axe of time is much helpful for readers because the bottom panel of Figure 4B includes several choruses in the same graph. The point of this simulation is that we succeed in reproducing both of the tri-phase synchronization and clustered anti-phase synchronization that were reported in our previous experiments of Ref. [12] (Note that we have changed the word “1:2 anti-phase synchronization” to “clustered anti-phase synchronization” as explained in our response to the comment of the other reviewer). In this revision, we added two enlarged graphs that show the process toward phase locking at clustered anti-phase synchronization and tri-phase synchronization, respectively (see Figure S1A and S1B in Supplementary Information). Note that, before and after the chorus of three frogs, there are short periods in which only two frogs make a chorus; in those period, a phase difference converges to π (i.e, anti-phase synchronization of two frogs is realized).

Comment 3

(Page 7, line 184)

The authors termed the function as “Dirac delta function”. It is, however, this is not accurate expression because $\delta = \infty$ when $\theta = 0$ in Dirac delta function. On the other hand their definition is $\delta = 1$ when $\theta = 0$, $\delta = 0$ otherwise.

Response:

Thank you for the comment. In this study, we assume a delta function $\delta(\theta_n(t))$ that is infinity at $\theta_n = 0$ and is 0 otherwise. Then, we also assume that $\delta(\theta_n(t))$ satisfies the relationship $\int_{t_{n,i}-\epsilon}^{t_{n,i}+\epsilon} \delta(\theta_n(t)) dt = 1$ for the call timing $t = t_{n,i}$ (Here, i represents the index of calls produced by the n th frog, and ϵ is a small positive parameter). The purpose of this assumption is to

propose a time differential equation in which the variables of E_n and T_n change discretely in time when the focal frog produces a call at $\theta_n = 0$. This delta function gives the deviations of T_n and E_n at the call timing $t = t_{n,i}$ as $\Delta T_n = \int_{t_{n,i}-\epsilon}^{t_{n,i}+\epsilon} \frac{dT_n}{dt} dt = \int_{t_{n,i}-\epsilon}^{t_{n,i}+\epsilon} \delta(\theta_n(t)) dt = 1$ and $\Delta E_n = \int_{t_{n,i}-\epsilon}^{t_{n,i}+\epsilon} \frac{dE_n}{dt} dt = \int_{t_{n,i}-\epsilon}^{t_{n,i}+\epsilon} (-\delta(\theta_n(t))) dt = -1$. Consequently, our model can describe the situation in which T_n is incremented by 1 and E_n is decremented by 1 every time the phase θ_n hits 0. In the revised manuscript, we added the explanation of the delta function $\delta(\theta_n)$ ([Lines 187-190 in Page 7, and Lines 195-200 in Pages 7 and 8](#)).

However, the delta function assumed here is slightly different from the Dirac delta function $\delta_{\text{Dirac}}(\theta_n(t))$. This is because the Dirac delta function satisfies the relationship $\int_{x_0-\epsilon}^{x_0+\epsilon} \delta_{\text{Dirac}}(x - x_0) dx = 1$ and also satisfies the relationship $\delta_{\text{Dirac}}(f(x)) = \frac{1}{f'(x=x_0)} \delta_{\text{Dirac}}(x - x_0)$ if $f(x)$ is zero at x_0 . If we use the Dirac delta function in our model, the deviation of E_n at $t = t_{n,i}$ is given by $\Delta E_n = \int_{t_{n,i}-\epsilon}^{t_{n,i}+\epsilon} \frac{dE_n}{dt} dt = \int_{t_{n,i}-\epsilon}^{t_{n,i}+\epsilon} (-\delta_{\text{Dirac}}(\theta_n(t))) dt = \int_{t_{n,i}-\epsilon}^{t_{n,i}+\epsilon} (-\frac{1}{\theta'_n(t=t_{n,i})} \delta_{\text{Dirac}}(t - t_{n,i})) dt = -\frac{1}{\theta'_n(t=t_{n,i})} \int_{t_{n,i}-\epsilon}^{t_{n,i}+\epsilon} \delta_{\text{Dirac}}(t - t_{n,i}) dt = -\frac{1}{\theta'_n(t=t_{n,i})}$, which is different from the result obtained from the above delta function. Therefore, we do not use the word “the Dirac delta function” in the revised manuscript. Moreover, we briefly mention that the delta function assumed here is different from the Dirac delta function, so as to avoid the confusion of readers ([Lines 190-192 in Page 7](#)). Thank you very much again for the valuable comment.